Curriculum Policy

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Introduction

Curriculum policy describes how states, districts, and schools manage the selection and use of instructional materials in America's K-12 public schools. The exact definition of "curriculum" varies—it can be used narrowly to refer only to textbooks. Usually, however, the term refers more broadly to "the program of instruction and sequence of experiences—and related resources (like lessons, activities, units, and textbooks)—that school districts use to ensure students master the academic standards each year in their coursework."[1] Today, curriculum materials are often software based and accessed online. Curriculum materials are not standards, which establish the skills and knowledge that students are expected to master in a given academic year but do not provide the content knowledge or suggest the teaching methods through which students will reach a particular standard.[2] Hence, teachers simply cannot teach standards by themselves—instead, standards must be embedded in instructional materials if they are to impact classroom practice.

The key argument for controlling or influencing the content of instructional materials is that the use of a strong published curriculum produces higher learning outcomes than would otherwise be produced if teachers curated and created their own materials. This evidence comes partly from international practice—countries with strong academic outcomes tend to specify specific curricular content[3]—and a nascent but growing body of research here in the United States.[4] Studies of specific curricular impact show positive results based on specific curricular choices.[5]

This argument is bolstered by the high mobility rate of America's K-12 student population.[6] Students move from school to school, where each curriculum in a given subject is often different, resulting in a fragmented learning experience in which key topics may be omitted while others are frequently repeated.

Finally, as detailed under key finding #3 below, when given the opportunity, a high percentage of teachers choose instructional materials that are below grade level, most especially with respect to underprivileged children, who are often children of color.

The counterargument—that curricular choices are best left to individual teachers—has strong support among teachers.[7] This view is reinforced in America's schools of education, which heavily favor the constructivist pedagogical views of John Dewey, namely, that teachers should build instruction around the interests and passions of their students.[8] It follows that imposing a single curriculum on children is viewed as wrong-headed from the start.

A second argument against externally imposed curricula, specifically in reading and social studies, focuses on their content. Liberal-minded critics assail religious content,[9] while conservatives voice concerns over teaching about the impact of racial discrimination.[10] A survey conducted by the National Education Association showed that "Almost half of schools reported challenges to teaching about issues of race and racism and policies and practices related to LGBTQ student rights. One-third reported attempts to limit student access to books in the school library and social and emotional learning."[11] Debates now rage across a wide spectrum of issues, from the inclusion (or exclusion) of specific materials and books in school libraries to the degree of multiculturalism (critics tend to prefer the term "identity politics") included in a specific curriculum.[12] Placing the choice of curricular content at the most local level absolves states of the responsibility to resolve these contested issues.

Finally, critics of the imposition of a curriculum designed outside of the classroom can also point to mixed evidence that these reforms work. For instance, the most extensive study of the use of new high-quality mathematics curricula found no resulting impact on students' academic scores,[13] and we currently lack strong evidence from cross-state studies that these reforms boost achievement.

I conclude with a discussion of understudied areas, including the role of school principals, effects on disadvantaged students and students who are below grade level, and the importance of system integration.

Key findings

Key finding #1: *Cross-country analyses of practice and early research on U.S. states provide support for the use of strong curriculum materials*.

There is a positive correlation between strong academic results and a country's use of a specific, content-rich curriculum. Even more compelling is the evidence that when a country relinquishes such a curriculum, the academic performance of students sharply declines. Research supporting the importance of curricular choice in the United States is especially strong in regard to literacy. After two decades of debate, the evidence supporting "the science of reading"—and the curriculum that embodies that science—is compelling. Most states and districts are currently moving to implement such curricula.

Key finding #2: *The results of recent research on the impact of the district-level adoption of strong materials have been somewhat positive but more mixed than the results of cross-national and cross-state studies.*

A review of the research record in the United States suggests that a strong curriculum can make a difference; however, in many cases, it has not done so.

Key finding #3: *Teachers' implementation of new, high-quality instructional material (HQIM) curricula has been very uneven.*

Most teachers are not using the new, high-quality curricula for most of their intended purposes. In fact, they continue to mix these curricula with multiple materials taken from the internet, and most of these materials are pitched below grade level.

Key finding #4: *To make good on the positive potential of a strong curriculum, research points to the importance of curriculum-specific professional learning (formerly called professional development) for teachers.*

Historical context and background

In the 19th century, the United States had a de facto national curriculum, the McGuffey Readers, which fell out of use in the early decades of the last century. However, controversy over the choice of instructional materials has been a feature of K-12 education at least since the post-Civil War period and continues to this day, with the so-called "culture wars" at the district and state levels. Pressure to control the content of the curriculum has come from both the left and right wings of the American political spectrum.

Allow me to offer a brief recap. From the Reconstruction onwards, former Confederate states issued guidelines for school materials that reflected their preferred narrative of the Civil War. Gradually, these guidelines were formalized by some states in the form of lists of approved textbooks. Today, there are nineteen states with state-appointed boards that review instructional materials and publish lists of "adopted" or "approved" textbooks for districts to consider. Even in these states, there is still substantial freedom for teachers to adopt "off-list" materials. In the thirty-one other states, school districts can choose instructional materials based on their best judgment.[14]

Two further developments—one federal and one largely state based—have impacted contemporary curriculum policy. Since the 1990s, under both Republican and Democratic presidencies, both Congress and the U.S. Department of Education (DOE) have moved to shift K-12 education policy from ensuring compliance with a set of educational inputs to holding schools accountable for educational outcomes based on a set of learning standards or skills in core academic subjects (English language arts (ELA), mathematics, and, to a lesser degree, science).

From 1990, when President Bush embraced outcomes-based education, to No Child Left Behind (NCLB) in 2001, Race to the Top (RTTP) in 2009, and the Every Student Succeeds Act (ESSA) in 2015, the U.S. DOE pushed standards-based accountability onto the U.S. K-12 system. In particular, RTTT, which gave a decisive advantage to states that had adopted the Common Core State Standards (CCSS) in ELA and mathematics, had a major impact on curriculum policy at the state level. In response to this grant opportunity, New York's commissioner at the time (this author) successfully sought funding for what was published as EngageNY, the nation's first high-quality, free, online, standards-based curriculum in mathematics and ELA.[15] Initially intended for use in New York state (NYS), this curriculum spread rapidly. RAND figures show that within a few years, it had been downloaded over 100 million times.[16]

Since then, publishers have increasingly embraced what is now called HQIM. Originally defined by EdReports as being aligned with the CCSS, the definition of HQIM has broadened to include rich and rigorous content, evidence-based practices, a full set of teacher supports, and embedded assessments, including diagnostic assessments in mathematics curricula.[17]

States have adopted very different policies towards HQIM, with at least a dozen, such as Louisiana and Tennessee, moving strongly to ensure adoption by all districts. In Louisiana, by 2017, over 80% of all public-school teachers were using the same ELA curriculum, Guidebooks.[18] The use of a single mathematics curriculum, Eureka, has reached approximately 90%.[19] In Tennessee, the percentage of teachers using HQIMs is now well over 90%.[20]

In turn, school districts have adopted very different curriculum policies. Most districts have adopted specific curricula in the major academic subjects, but some—including New York City Public Schools—have, until very recently, left the choice to individual schools and teachers.[21] An increasing number of districts are trying to ensure that teachers use *only* the district-selected curriculum, but data from RAND show that, recently, some 90% of American public-school teachers were mixing their district's curriculum with materials drawn from the internet.[22]

Evidence supporting key findings

Key finding #1: *Cross-country analyses of practice and early research on U.S. states provide support for the use of strong curriculum materials.*

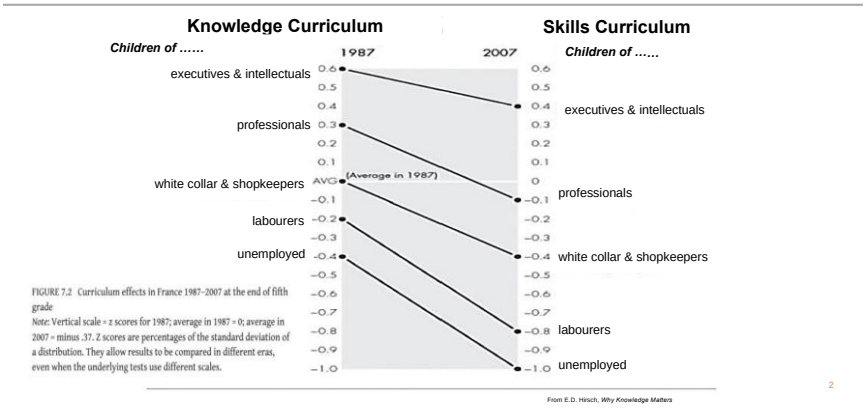
There is a positive correlation between strong academic results and a country's use of a specific, content-rich curriculum. Even more compelling is the evidence that when a country relinquishes such a curriculum, the academic performance of students sharply declines.

As Ashley Berner and I wrote in "Hiding in Plain Sight,"

> The most extensive study to date found that a comprehensive, content-rich curriculum was the salient feature in nine of the world's highest-performing school systems as measured by the Programme for International Student Assessment (PISA). Despite the vast cultural, demographic, political, and geographic diversity of Finland, Hong Kong, South Korea, Canada, Japan, New Zealand, Australia, the Netherlands, and Switzerland, their educational systems all emphasized content-rich curriculum and commensurate standards and assessments.[23]

While this evidence is only correlational—we lack causal research that links a national or state-level high-quality curriculum to national educational outcomes—it is further strengthened by negative evidence. When France dropped its national curriculum in favor of allowing teachers to create their instructional content, students performed steadily worse in reading, with the greatest damage done to the most disadvantaged students:[24]

Curriculum Effects: France Abandons a Content-Rich National Curriculum



Note: Reprinted with permission from Why Knowledge Matters: Rescuing Our Children from Failed Educational Theories (p. 145), by E. D. Hirsch, 2016, Harvard Education Press.

The same negative consequences of abandoning a mandated, specific curriculum have impacted both Finland and Sweden.[25] When nations do the reverse—that is, introduce rigorous, knowledge-rich curricula that are assessed in national tests, as England did in 2014—student outcomes rise.[26]

For many years, the Organisation for Economic Co-operation and Development (OECD) has maintained that curriculum-centered instruction should be considered a suboptimal method of pedagogy—it prefers some version of child-centered learning based on materials picked by autonomous teachers choosing their preferred instructional content. Remarkably, the data from the PISA assessment, which the OECD designs and implements, show just the reverse: strong results are achieved when a nation's teachers use a strong, shared curriculum supported by curriculum-aligned assessments.[27]

The evidence for the positive impact of specific learning materials is strongest in regard to learning to read. At least since the publication of the Report of the National Reading Panel in 2000, policymakers have had powerful evidence that their current models of literacy instruction, known as "whole-language" or "balanced literacy," were deficient.[28] That research report showed that an approach that focused on phonics, phonemic awareness, fluency, vocabulary, and knowledge-building was the most effective way to teach students to read. However, it has only been in the last five years or so that states have embraced this approach, widely known as the science of reading.

It was Mississippi, under the leadership of State Superintendent of Education Carey Wright, that first adopted statewide instructional materials in the science of reading.[29] A multifaceted effort to provide teachers with new curricula and the professional learning required to use them effectively produced outsized gains in reading proficiency. Mississippi went from being ranked 49th in 4th-grade reading proficiency in 2013 to being ranked 21st in 2022.[30]

Importantly, these results were not due to curriculum use alone. For example, a state law that enabled the retention of 3rd-grade students with weak reading skills was a critically important component. Perhaps more importantly, the gains have been most pronounced

among elementary-school children. Eighth-grade results have not shown the same jumps.[31] One possible explanation is that while policymakers have focused on phonics instruction, one key element of the science of reading, namely, the building of background knowledge, has not, until very recently, been at the core of the reforms in literacy instructional materials.

The idea is simple but central: if students do not know about the subject matter of a given piece of writing, they will not be able to understand it. The importance of using a curriculum that enables students to build extensive knowledge through reading materials that encompass different genres of literature, social studies, science, and the arts has been demonstrated repeatedly in the research cited by E. D. Hirsch and, more recently by Natalie Wexler and Dan Willingham.[32] As Willingham notes, "The mistaken idea that reading is a skill may be the single biggest factor holding back reading achievement in the country. The knowledge base problem must be solved."[33]

Starting with the Core Knowledge curriculum pioneered by Hirsch[34] and the International Baccalaureate (IB), widely used HQIM materials in ELA (such as Guidebooks, Wit & Wisdom, and EL Education) are now explicitly focused on knowledge-building. State standards are also becoming more explicit about defining the knowledge that students should master. Recently, Texas announced a new set of ELA standards that do just that.[35]

Key finding #2: *The results of recent research on the impact of the district-level adoption of strong materials have been somewhat positive but more mixed than the results of cross-national and cross-state studies.*

A review of the research record in the United States suggests that a strong curriculum can make a difference; however, in many cases, it has not done so.

American-based research on the impact of curricular choices on student learning outcomes is relatively recent. Early research up to the turn of this century focused on assessing the impact of students' prior knowledge on assessment outcomes, but with rare exceptions, it did not focus on a specific curriculum.[36] The first papers to argue explicitly for the importance of curricular choices, published in 2009 and 2012, could point to a small group of studies on specific curriculum effects, but they did not yet have a robust set of primary research findings to draw from.[37]

In the last quarter century, several further studies have strengthened our knowledge base. In 2010, researchers conducted a randomized controlled trial (RCT; considered the "gold standard" of research) of the impact of specific mathematics curricula on student learning outcomes.[38] Two further RCTs followed, each of which evaluated the effectiveness of a single textbook.[39] In addition to these RCT studies, there has been a series of correlational studies that have reviewed the use of multiple curricula in mathematics.[40]

Taken together, these studies tend to show a positive impact from switching to new, stronger materials by contrast to what is known as "business as usual," which, in most cases, involved the use of materials that were not standards-aligned and/or were created by teachers or their local districts. This is evident in one extensive summary that found that curriculum shifts produced an effect size of +0.05 to +0.10 standard deviations (s.d.).[41] (An effect size of .07 s.d. would mean approximately 35 days of learning for a fifth- or sixth-grade student in reading and slightly fewer days in mathematics).

However, these studies of positive results involved a relatively small number of teachers and schools. We lack robust studies of large-scale shifts: reference was made above to one study of the impact of new mathematics curricula in multiple states—a study that showed no impact.[42] To date, there is no consistent evidence from cross-state studies that HQIM implementation improves student outcomes on the National Assessment of Educational Progress (NAEP).[43]

The next key finding focuses on why we are seeing so little overall impact from the new curriculum materials.

Key finding #3: *Teachers' implementation of new, HQIM curricula has been very uneven.*

Most teachers are not using the new, high-quality curricula for most of their intended purposes. In fact, they continue to mix these curricula with multiple materials taken from the internet, and most of these materials are pitched below grade level.

An extensive 2020 review of teachers' use of new mathematics curricula showed that only "25% of teachers use the textbook in nearly all their lessons for all essential activities, including in-class exercises, practice problems, and homework problems."[44] Survey data from 2019 showed that teachers are mixing their district-mandated curricula with a variety of internet-sourced content: "The percentage of principals who reported providing teachers with standards-aligned curricula was much higher than the percentage of teachers who reported using standards-aligned curricula for the majority of their instructional time."[45] That same survey found that 85% of teachers use supplemental digital materials to plan their instruction and that 88% use such materials to teach their classes.[46]

Further survey data have been used to explore why teachers are choosing to mix multiple kinds of instructional materials. A 2018 review of teacher practice across the United States found that 80% of the material that teachers were pulling from the internet for use in their instruction was below grade level.[47] A 2023 survey report found that "Three in ten K-12 teachers consider the curriculum materials that are required or recommended by their schools or districts to be too challenging for the majority of their students."[48]

Finally, my 2023 multidistrict study of middle-school ELA and mathematics classrooms found that even when teachers were referencing their district-adopted HQIM curriculum materials, the majority used methods to water down the rigor of the curricular content. These methods included teachers rapidly answering questions that they asked of students and simplifying these questions.[49]

One reasonable inference from these studies is that teachers do not believe that their students can learn from the more rigorous HQIMs that their districts are increasingly asking them to use.

Key finding #4: *To make good on the positive potential of a strong curriculum, research points to the importance of curriculum-specific professional development for teachers.*

Studies cited under key finding #2 have shown that switching instructional materials *can* have a positive impact on student learning outcomes. A recent experimental design study of a group of charter schools that used Core Knowledge HQIMs showed statistically significant, positive outcomes for the students whose teachers used these materials.[50]

If HQIMs can make a difference, why is it the case that they often do not? The comprehensive research on mathematics adoptions cited earlier—which showed no learning impact—provides us with some reasonable hypotheses. The authors of that study found that, on average, districts implementing a new mathematics curriculum offered teachers just over a single day of professional development each year. In the context of the average American teacher, who has some fourteen years of teaching experience, such a small dosage of information and guidance clearly is not enough to create widespread adoption of the new materials.[51]

Other studies reinforce these findings. A second study examined 95 experimental and quasi-experimental preK-12 science, technology, engineering, and mathematics (STEM) curriculum-related professional development programs. While it found a positive overall impact of the programs (0.21 s.d.), the study concluded that "Programs saw stronger outcomes when they helped teachers learn to use curriculum materials; focused on improving teachers' content knowledge, pedagogical content knowledge and/or understanding of how students learn; incorporated summer workshops; and included teacher meetings to troubleshoot and discuss classroom implementation."[52] Another survey study reported mixed results on the impact of new curricula on teacher practice, and it concluded that

Teachers are better positioned to use curricula well in their classroom lessons if they work within an instructional system in their school and district where standards, curricula, assessments, professional learning, and evaluation are all in alignment and working together to help teachers engage students in standards-aligned classroom practices.[53]

An RCT study of HQIM mathematics impact found that teachers who received professional support for their implementation of the new curriculum produced higher learning outcomes than did those who were only given the new materials.[54]

While professional learning (PL) can help, research suggests that it cannot by itself transform HQIM usage rates or the fidelity of HQIM implementation. The RCT study cited in the previous paragraph did find PL made a difference—but that difference was a modest 0.03 s.d. There is some evidence that PL that is explicitly linked to supporting a teacher's familiarity with a specific curriculum and that is provided over multiple dosages can be effective.[55] Research also suggests that PL is more effective when teachers have the chance to practice specific skills identified in their coaching collectively.[56] The reason that the researchers found positive impacts of the use of Core Knowledge curricular materials in charter schools may be related to the extensive professional learning, strong learning culture, and strong principal support for the use of this curriculum in these schools.

What we do not know

- School Principals: There is considerable evidence that principals are important for creating high-performing schools.[57] It may well be true that this is true with respect to the implementation of new curricula such as HQIMs—and education reform organizations routinely make this point: "If school leaders and their leadership team members do not understand the curriculum deeply, they will not be effective in supporting teachers to do the same."[58] Common sense suggests that when principals, for example, include curriculum-integrated rubrics in their classroom observations, they promote teachers' fidelity of curriculum implementation. However, we have no published research that supports this claim.

- Underprivileged Students: While there is some evidence that providing more rigorous instructional materials to students of color raises their learning levels,[59] to date, research on the impact of HQIMs has largely focused on the impact of switching to a stronger curriculum *on the average student*. However, a great deal of education policy is focused on raising the educational outcomes of underprivileged students. Hence, assessing the impact of switching to HQIMs on this student population is a key concern. We know almost nothing about this topic. Reports such as the well-known "Opportunity Myth" from TNTP (formerly known as The New Teacher Project) stress that underprivileged students—especially if they are of color—are taught "down to," meaning that teachers dilute and simplify instructional materials, with the result that such students are often never taught content at their grade level.[60] The implication is that if these students were taught more rigorous, grade-level material, their outcomes would be higher. To date, however, we cannot conclusively prove that this is the case. At most, we can infer from the results of public schools run by the strongest urban charter management organizations that rigorous content is probably an important part of the many policies and practices that such organizations have put in place.[61] However, efforts to accelerate a reversal of the learning losses of underprivileged students in noncharter public schools by trying to support their access to and use of grade-level HQIMs have not yet achieved strong outcomes.[62]

- Learning Gaps: There is much that we currently do not know about the potential impact of the new curricula on students who have unfinished learning. Allow me to pose two questions. First, is exposure to grade-level HQIMs good for all students, no matter how far behind they are, or is such exposure more effective for students who are, say, within one year of being at grade level in their knowledge and skills? Second, for different levels of gaps in previous learning, what kinds of diagnostic data and pedagogical support can best leverage the contribution of HQIMs to student learning?

- System Integration: To date, public education in the United States is largely siloed: teacher preparation, curriculum use, and testing are not interconnected.[63] It may well be that the impact of HQIMs is muted because schools of education do not prepare teachers to teach HQIMs and because states do not test what students study in the HQIMs that their school districts select. However, at this point, we can only speculate.

Endnotes and references

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